ChatGPT

Executive Summary

Orbya will be a Nepal-focused travel metasearch and booking portal (flights, hotels, car rentals, taxis, guides, treks) with an AI-driven trip planner. To launch fast with minimal funding, we'll follow an **affiliate-first / hybrid approach**: integrate existing travel APIs (Booking.com, Amadeus/Skyscanner, Expedia, etc.) for global inventory and build our own inventory for local services (taxis, guides, adventures). The architecture will use a modern **JavaScript stack** (React/Next.js front-end; Node.js/Express backend; PostgreSQL DB; Redis cache) deployed on a generic PaaS ("Antigravity"). Mapbox GL JS (with optional Three.js) will power interactive 2D/3D maps [1] [2]. We'll incorporate OAuth/JWT authentication and an admin panel for management. Critically, Orbya's **AI Trip Planner** will call a GPT-4 style LLM (e.g. OpenAI) with contextual prompts and travel data to generate day-by-day itineraries, similar to Booking.com's AI planner [3] [4].

The 7-day rapid-build plan (assuming a small team or solo founder with Antigravity) is to scaffold the app and deliver a minimal MVP: Day 1: environment, auth, basic UI; Day 2–3: hotel search integration (Booking.com affiliate) and listing UI; Day 4: taxi booking flow; Day 5: guide listings and car-rental API (e.g. Booking/CJ or DiscoverCars affiliate); Day 6: flights integration (Skyscanner or Amadeus) and bus info; Day 7: AI trip planner, payments, admin panel. (See Timeline table below.)

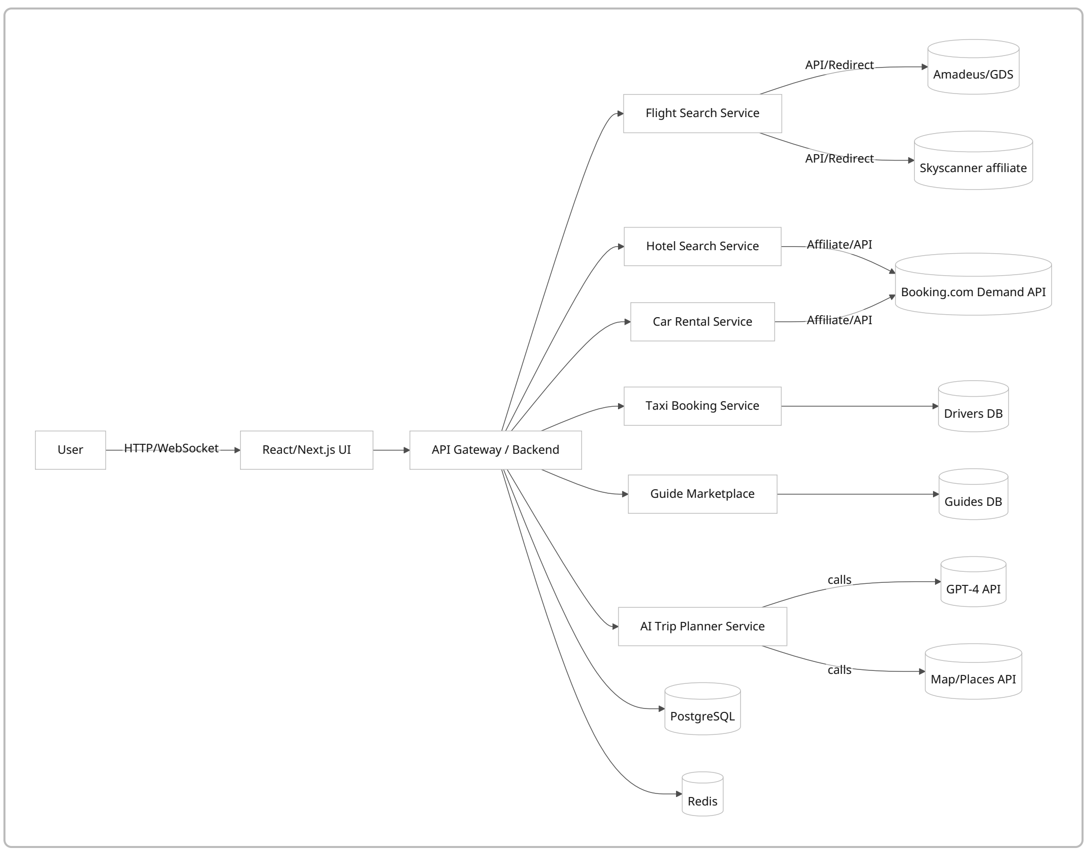
Orbya's monetization combines **affiliate/CPC and commissions**: we earn per-click or per-booking from partners (Booking.com ~4–6% [5], Skyscanner ~$0.40–$1 per flight click [6]), plus direct 10–20% commissions on our own bookings (taxis, guides, treks, local hotels). A premium subscription for unlimited AI itineraries adds recurring revenue. We'll implement analytics (Google Analytics/Pixel, or PostHog) to track usage and conversions. Security best practices (HTTPS, input sanitization, OAuth) and tests (unit, integration, load) will be applied. The following sections detail the tech stack, system design, data flows, API integrations, UI/UX components, database schema, monetization math, and a 7-day development schedule.

Tech Stack & Architecture

Layer	Technology	Notes/Alternatives
Frontend	React + Next.js, Tailwind CSS	Server-side rendering, fast prototyping
Backend	Node.js (Express or NestJS)	JavaScript/TypeScript, easily integrates APIs
Database	PostgreSQL	Relational for structured bookings, users
Caching	Redis	Cache API results, sessions

Layer	Technology	Notes/Alternatives
Search/ Index	Algolia (optional)	Fast search/filtering (flights/hotels)
Auth	Firebase Auth or Auth0 (OAuth/JWT)	Social logins (Google/FB), JWT for APIs
Maps/3D	Mapbox GL JS (3D terrain, place search) [1], Three.js	Rich map UI, 3D models [2]
AI/LLM	OpenAI GPT-4 API (or Azure OpenAI)	Itinerary generation (AI Trip Planner)
Hosting/ PaaS	Antigravity (generic PaaS), Vercel (frontend)	Rapid deployment, scaling
API Gateway	Built-in (Next.js API routes or Node Express)	Route traffic to services, caching
CI/CD	GitHub Actions, Railway.dev, Vercel	Automated builds, deploy on push
Payment	Stripe or PayPal (with Nepali support)	Booking payments, subscription billing
Analytics	Google Analytics, PostHog, Segment	User behavior, funnels, A/B test tracking

System Architecture (conceptual):



Key points:

- All user requests go through our frontend, which calls backend APIs via GraphQL or REST. The API layer routes to microservices for each function.
- **Flight Search:** Initially, use flight metasearch/APIs (Amadeus Self-Service API covers 400+ airlines including low-cost [7], or Skyscanner affiliate links [6]). For domestic Nepal routes, also consider direct airline partners (Buddha Air, Yeti Air).
- **Hotel Search:** Use Booking.com Demand API (affiliate) or Expedia Rapid API for hotel inventory. These support real-time availability and can return properties via our backend [8].
- **Car Rentals:** Booking.com's Cars API (8.2% commission per CJ) or affiliate programs (DiscoverCars pays ~$20 per booking [9]). Travelpayouts lists many (Rentalcars.com 6% on CJ [10]).
- **Taxi/Bike Rental:** Build our own booking system. Drivers and vehicles register via an app; customers book routes (e.g. from/to airport). Store drivers/vehicles in DB, route requests to available drivers.
- **Tourist Guides:** Similar marketplace. Guides sign up with profiles; tourists request days. Manage bookings/comms.
- **Bus/Transport:** We can start with static routes and calls (e.g. Nepal Yatayat) or partner (KTMCTC bus API if any).

Caching (Redis) will store recent search results to reduce API calls. Use a CDN (e.g. Vercel) for static assets. All services communicate over HTTPS with appropriate API keys.

Third-Party APIs & Integrations

Orbya will integrate **affiliates and APIs** for travel content:

- **Flights:**
- *Amadeus Self-Service APIs:* Provides flight offers and booking for 400+ airlines [7]. Offers rich data but requires registration and per-call charges.
- *Travelport/GDS:* (Optional, for airline ticketing integration; complex to set up).
- *Skyscanner Affiliate:* Use Skyscanner's affiliate widgets or Flights API (via Impact network) for prices [6]. Skyscanner is CPC-based (~$0.40–$1 per click for quality traffic [6]).

- *Direct Airlines:* Provide links to Nepal airlines (Buddha Air, Yeti Air, Shree Air). Embed affiliate codes if available.

- **Hotels:**

- *Booking.com Demand API:* As an affiliate, fetch properties by JSON queries (see example below) [8]. The API returns hotel listings, rates, and availability. Booking.com pays ~4–6% commission on stays [5].
- *Expedia Rapid API:* Similar hotel inventory with connectivity.

- *Connectivity API:* Eventually, onboard local hotels directly into our DB and API (gain 10–20% commission margin vs. ~4% affiliate).

- **Car Rentals:**

- *Booking.com Cars API:* Access car rental inventory via Booking.com (6% commission via affiliate [5]).
- *DiscoverCars Affiliate:* High payout (~$20/booking [9]). Could integrate via XML/affiliate.
- *Rentalcars.com:* (Via CJ, ~6% commission [10]).

- **Taxi/Bike Services:**

- Build an in-house service. No standard API; we'll create endpoints for drivers (register, accept jobs). Use Google Maps/Mapbox Geocoding APIs for address lookup. Payment can be collected through our gateway and we disburse to drivers.

- **Tourist Guides & Tours:**

- Also in-house. Guides upload profiles (languages, expertise). Bookings create an assignment. Optionally integrate local travel agencies or Airtasker-like services.

- **Maps/Geo:**

- *Mapbox GL JS:* 2D/3D maps with satellite and terrain [1]. Mapbox supports 3D terrain and can use custom Three.js layers for models [2]. Useful to plot routes (e.g. trekking paths) and landmarks (e.g. mountain peaks).
- *Google Maps API:* Geocoding and Places (location autocomplete, POIs).

- **Payment Gateway:**

- *Stripe/PayPal:* Accept international payments. For Nepali users, consider local gateways (eSewa, Khalti) or cash-on-delivery options. Ensure PCI compliance.

- **AI & Data:**

- *OpenAI GPT-4 API:* For the AI Trip Planner. Prompt with user inputs (budget, days, interests) to generate itinerary text. Possibly chain with Google Knowledge/Maps for specific location data (like addresses). Cache popular itineraries in DB to reuse.

- **Search:**

- Optionally, use a search/index service like Algolia or Elasticsearch to index hotels, guides, places for fast querying (especially if many listings).

Sample API Integration Snippet

Booking.com Hotel Search (Node.js example):

```
const axios = require('axios');
async function searchHotels(cityId, checkin, checkout, adults) {
```

```
  const query = {
    city: cityId,
    checkin, checkout,
    guests: { number_of_rooms: 1, number_of_adults: adults }
  };
  const res = await axios.post('https://distribution-xml.booking.com/json/v1/
search',
    query,
    { headers: { 'Authorization': 'Bearer <API_TOKEN>', 'X-Affiliate-Id':
'<AFFILIATE_ID>' } }
  );
  return res.data.results;
}
```

This uses Booking.com's Demand API (JSON) with affiliate credentials. The request format matches the example in the docs [8] .

Skyscanner Flight Search (REST pseudocode):

```
// (Assuming access to a Skyscanner Partner API or redirect link)
const flightResults = await axios.get(`https://partners.api.skyscanner.net/
apiservices/pricing/v1.0`, {
  params: { origin: 'KTM', dest: 'PKR', date: '2026-07-15', ... }
});
```

OpenAI Itinerary Prompt (Node.js):

```
const { Configuration, OpenAIApi } = require('openai');
const openai = new OpenAIApi(new Configuration({ apiKey:
process.env.OPENAI_API_KEY }));

async function generateItinerary(destination, days, budget, category) {
  const userPrompt = `Plan a ${days}-day ${category} trip to ${destination} for
1 person with a budget of NPR ${budget}. Include flights, hotels, daily
activities.`;
  const res = await openai.createChatCompletion({
    model: "gpt-4",
    messages: [{role: "user", content: userPrompt}]
  });
  return res.data.choices[0].message.content;
}
```

This mirrors Booking.com's approach of asking open-ended queries like "Where should I go for a romantic weekend?" and getting full itineraries [3] .

Integration Patterns

- **Affiliate (Booking.com/CJ):** Use backend calls to affiliate APIs or embed partner widgets. All booking links include our affiliate tracking. Commission rates start ~4% for hotels and 6% for car rentals [5].
- **Redirection:** For flights, we may show search results and then redirect the user to partner booking pages (airline or OTA) with our affiliate tag, following a metasearch model [6].
- **Direct Booking:** For services we own (taxis, guides, local hotels), handle end-to-end: take payment, store booking in our DB, and notify suppliers.

AI Trip Planner Architecture

Orbya's AI feature will be a multi-stage pipeline:

1. **User Input:** Budget, days, people, preferences (style: Budget/Comfort/Premium/Luxury).
2. **Prompt Engine:** Construct a GPT-4 prompt combining these inputs with static info about Nepal destinations. Example: "I have NPR 50,000 for 5 days in Kathmandu and Pokhara, traveling with family, prefer comfort." Use few-shot examples to guide output structure.
3. **LLM Call:** Invoke the GPT-4 API with this prompt. As Booking.com did, this can be done within weeks [11].
4. **Parse Response:** Extract key itinerary items (day 1: flight, hotel, activity; day 2: …). Possibly use NLP or a standardized JSON output schema.
5. **Data Enrichment:** For each place/activity mentioned, fetch details (coordinates, photos) via Google Places or Mapbox and attach to itinerary (for map plotting).
6. **Cache/Store:** Save the itinerary (query + result) in our DB so repeat queries are instant.
7. **UI Presentation:** Display day-by-day plan on UI, with map and cost breakdown.

Scalability: Cache common plans. Add prompt caching. Limit free AI queries (e.g. 3 free), with subscription for unlimited.

Example Flow: User requests 7-day Premium trip. Backend hits OpenAI with prompt. OpenAI returns a 7-day plan referencing flights from KTM to POK, a 4-star hotel, Sigiri Lake tour, etc. Orbya maps each item to actual listings/prices (via our services/APIs) and shows a consolidated plan. This mimics Booking.com's AI planner (built in 10 weeks, combining GPT with their data) [11] [3].

Database Schema (Sample)

Table	Fields
Users	id, name, email, password_hash, role (user/admin)
Hotels	id, name, location (lat,lon), address, amenities, rooms_available, base_price, affiliate_id
Flights	id, airline, flight_no, origin, dest, depart_time, arrive_time, price, currency
CarRentals	id, supplier, car_model, seats, price_per_day, location
Drivers	id, name, vehicle_type, vehicle_number, phone, rating

Table	Fields
Guides	id, name, languages, daily_rate, experience_years, bio
BusRoutes	id, company, origin, dest, schedule, price
Itineraries	id, user_id, title, total_cost, details (JSON)
Bookings	id, user_id, service_type, service_id, date, status, amount
Reviews	id, user_id, service_type, service_id, rating, comment
AIPlans	id, user_id, prompt_text, created_at, response_text

(Foreign keys: Bookings.service_id refers to id in Hotels/Flights/Drivers/etc.)

We'll use an ORM (Prisma or Sequelize) to define this schema. Each service (FlightsSvc, HotelsSvc, etc.) will read/write relevant tables (e.g. FlightsSvc queries Hotels and Bookings).

API Design

REST Endpoints (examples)

Method	Endpoint	Description	Auth
`POST`	`/api/auth/login`	Email/password login (returns JWT)	No
`GET`	`/api/flights/search`	Query flights (params: from, to, date)	No
`GET`	`/api/hotels/search`	Query hotels (city_id, checkin, checkout, guests)	No
`GET`	`/api/cars/search`	Query car rentals (location, dates)	No
`POST`	`/api/bookings`	Create a booking (user must be logged in)	Yes
`GET`	`/api/bookings`	List user's bookings	Yes
`POST`	`/api/taxis/request`	Request taxi (pickup/dropoff)	Yes
`POST`	`/api/guides/book`	Book guide (guide_id, days)	Yes
`POST`	`/api/ai/plan`	Generate AI itinerary (with preferences)	Yes
`GET`	`/api/admin/stats`	Site analytics (admin only)	Yes (admin)
`GET`	`/api/search/autocomplete`	Autosuggest places (calls Mapbox Geo)	No

We may also support GraphQL (Apollo Server) to flexibly combine searches.

Authentication & Admin

- **Authentication:** Use JWT tokens or Firebase Auth. Support social login (Google). Protect booking and planner endpoints.
- **Roles:** `user` vs. `admin`. Admin panel (e.g. Strapi or custom React admin) to manage inventory: add/edit hotels, approve guides, view reports.
- **Admin Dashboard:** Key metrics (bookings by category, revenue, popular searches). Possibly via a BI tool or embedded (Metabase/Redash).
- **User Dashboard:** View itinerary, bookings history, favorite destinations.

Security: Always use HTTPS. Sanitize inputs to prevent SQL/XSS. Validate tokens. CORS settings for API.

UI/UX & MVP Features

Core User Flows (MVP)

1. **Search & Book:** On homepage, users pick a service: Flights, Hotels, Cars, Taxis, Guides, Bus, or start an AI Plan. Show simple search forms with date pickers and filters (price, rating).
2. **Results:** Display results as cards/grids (images, summary). Example: flight list sorted by price; hotel list with price per night. Include map view button to see results on map.
3. **Detail/Booking:** Clicking a result leads to a detail page (hotel description, images, price breakdown). A "Book Now" button either processes via our checkout (if direct) or links to partner booking (with affiliate tag).
4. **Taxi/Guides:** User can book taxi from A to B by selecting vehicle type and time. Guide booking: view profiles, then request date(s).
5. **AI Trip Planner:** Wizard UI: step 1 gather trip parameters (destination, dates, budget, category), step 2 display AI-generated itinerary (editable), step 3 allow saving or booking items.

UI Components (wireframe ideas)

- **Navbar/Header:** Orbya logo; menu (Flights/Hotels/Taxis/Guides/Car-Rental/Bus/AI-Planner/Login).
- **Hero/Search Bar:** Prominent search fields (e.g. "Where to?" with date and guests for hotels).
- **Result Lists:** Filter sidebar (price range, star rating, facilities); main area listing results.
- **Map View:** A map panel (e.g. left or top) showing pins for results, interactively update list.
- **Trip Planner Page:** Cards for each day itinerary; map overlay of route; cost summary.
- **Footer:** Links (About, Terms), contact info.

(UI mockups: could adapt templates from travel site frameworks – e.g. use Visily Travel Template [12] or open-source dash components.)

Monetization Model

- **Affiliate Commission:**
- *Flights:* If using Skyscanner/Flight APIs, we earn CPC or per-sale commissions. Example: Skyscanner pays ~$0.4–$1 per click [6], or a percentage via Travelpayouts.
- *Hotels (Booking.com):* ~4% of booking value [5]. E.g. 100 bookings/month @ ₹8,000 avg yields ₹32,000/month (at 4%).

- *Car Rentals:* Booking.com 6% [5] or $15–20 from DiscoverCars [9].
- *Attractions:* ~4% (Booking) [5] (future expansion).
- **Direct Commission:**
- *Taxi:* We set commission (e.g. 15%) on fares. If a ride costs NPR 2,000, Orbya gets NPR 300.
- *Guides:* 15–25% on tour/guide fees. E.g. NPR 4,000/day guide = NPR 600–1,000 to Orbya.
- *Bus/Treks:* Commission on package price (10–20%). Trekking packages especially lucrative (15–25%). E.g. 10 pax trek @ NPR 30,000 = NPR 4,500 at 15%.
- **CPC Advertising:**
- Banner ads or promoted placements (e.g. hotels or tours can sponsor spots).
- **Subscription (AI Planner):**
- Free tier (e.g. 3 plans/month), Premium (unlimited, advanced options) at ~$5–10/month.
- Example: 500 premium users @$10 = $5,000/month.

Revenue Example (monthly):

```
100 hotel bookings @₹5,000 avg * 10% direct commission = ₹50,000
200 taxi rides @₹1,500 avg * 15% = ₹45,000
50 guide days @₹4,000 * 20% = ₹40,000
100 flight clicks @$0.5 = ₹4,250 (≈$50)
300 AI premium subs @$600/mo = ₹180,000
Total ≈ ₹319,250 (₹3.8M/year)
```

(Orbya margins improve as more local inventory is onboarded, moving away from low affiliate percentages.)

Deployment & Security

- **Deployment:** Use Antigravity PaaS (assumed to support Node/Next/DB). Alternatively, host frontend on Vercel, backend on Railway or Heroku, DB on Supabase. Ensure CI/CD pipelines push code on commit.
- **Environment:** Production domain (e.g. orbya.com), SSL certificate (Let's Encrypt). Config vars: API keys, DB credentials.
- **Performance:** Use CDNs for assets. Enable server-side rendering (Next.js) for SEO. Set cache headers for static data.
- **Security:** HTTPS only. Sanitize all inputs (prevent XSS, SQLi). Use OWASP guidelines. Secure JWT secrets. Rate-limit APIs to prevent abuse.
- **GDPR/Privacy:** Even if Nepal-based, comply with data protection (encrypt PII, privacy policy for user data).
- **Payments:** Use PCI-compliant checkout (Stripe Elements or PayPal).

Testing & Analytics

- **Unit/Integration Tests:**
- Backend: Jest or Mocha to test API endpoints (search, booking creation).
- Frontend: Cypress for UI flows (search → results → booking).
- **Load Testing:** Use tools like k6 or Locust to simulate search traffic (especially flights/hotels searches).

- **Monitoring:** Log errors (Sentry), performance metrics (Datadog/New Relic).
- **Analytics:** Google Analytics + Facebook Pixel for marketing, or self-hosted (PostHog) for event tracking (click-through, bookings, planner usage). Use funnels to optimize conversion.

7-Day Build Timeline

Day	Activities	Deliverables
1	*Setup:* Project repo, PaaS environment, domain. Implement user auth (JWT/Firebase) and basic UI skeleton (landing page). Install frameworks (React/Next, Node).	Code scaffolding, Auth working, placeholder UI.
2	*Hotels MVP:* Integrate Booking.com Demand API (fetch listings by city/date). Build hotel search form and result list (demo data first). Implement filters (price, rating).	Hotel search page + results displayed (affiliate links).
3	*Taxi & Guides:* Create DB tables for Drivers/Guides. Develop signup flows for drivers/guides. Build taxi search form (pickup/dropoff). Show list of available drivers. Basic guide directory page.	Taxi/Guide listing pages (static data), driver signup form.
4	*Car Rentals:* Integrate Booking.com Car API or affiliate link. Add car rental search form and results. *Flights MVP:* Stub flight search using Skyscanner affiliate (redirect links).	Car rental search UI + links; Flight search form (redirects in place).
5	*AI Planner Backend:* Integrate OpenAI API. Build `/api/ai/plan` endpoint that takes user prefs and returns itinerary. Connect map (Mapbox) to UI to plot itinerary waypoints.	AI Trip Planner UI (input form) and initial GPT output displayed.
6	*Booking System:* Implement booking endpoint to record a selection (hotel/flight/taxi). Integrate Stripe for payments. Add basic user dashboard ("My Bookings").	Booking flow working; payment processing stub (test mode).
7	*Admin & Polish:* Build admin panel for managing listings (can use a template). Add analytics (Google Analytics). UI/UX polishing: responsive design, error handling. Final testing and deployment.	End-to-end MVP: login, search, book, plan. Deployed on Antigravity.

Each day includes testing of components. If solo, tasks may overlap or extend a bit beyond one day.

Citations

- Booking.com Affiliate API examples [8], commission tiers [5].
- Skyscanner affiliate insights [6], API docs 【4†】.
- Expedia Rapid API overview [13] (Car API stats).
- AI Trip Planner by Booking/OpenAI [3] [4].

- Mapbox GL JS (3D maps) [1], Three.js integration [2].
- Amadeus flight API coverage [7].

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[1] Create 3D and Dynamic Web Maps with Mapbox GL JS
https://www.mapbox.com/mapbox-gljs

[2] Add a 3D model with three.js | Mapbox GL JS | Mapbox
https://docs.mapbox.com/mapbox-gl-js/example/add-3d-model/

[3] [4] [11] Booking.com and OpenAI personalize travel at scale | OpenAI
https://openai.com/index/booking-com/

[5] Join Booking.com's travel affiliate programme | CJ
https://www.cj.com/en-gb/publisher/partners/booking.com

[6] Skyscanner Flight Affiliate Program | Travelpayouts
https://www.travelpayouts.com/blog/skyscanner-flight-affiliate-program/

[7] Amadeus Travel API
https://www.flightslogic.com/amadeus-travel-api.php

[8] Try our Demand API
https://developers.booking.com/demand/docs/getting-started/try-out-the-api

[9] [10] Car Rental Affiliate Program: Payouts $20+ Per Client | Discover Cars
https://www.discovercars.com/affiliate

[12] Travel Website Wireframe Template - Visily
https://www.visily.ai/templates/travel-website-wireframe/

[13] Rapid Car API for Scalable Car Rental Bookings | Expedia Group
https://partner.expediagroup.com/en-us/landing-pages/rapid-car-api-for-scalable-car-rental-bookings